CS 536 — Assignment 5: Collective Communication

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GitHub repository: https://github.com/arav-p/networks-assignment-5

1 Overview

This report covers custom implementations of five collective-communication algorithms over the PyTorch `torch.distributed` **gloo** backend, together with an experimental evaluation across message sizes and process counts.

Algorithms implemented (see `collectives.py`):

Collective	Algorithms
AllGather	Ring, Recursive Doubling, Swing
Broadcast	Binary Tree, Binomial Tree

Every algorithm is built from the point-to-point primitives `dist.send`, `dist.recv`, `dist.isend`, `dist.irecv` only — the built-in collectives are never used inside the algorithms under test.

2 Algorithm descriptions

2.1 AllGather

Each of P processes starts with one chunk of size `C`. After the collective, every process holds all P chunks in rank order.

Ring — $P - 1$ rounds. At round s every rank sends the slot it just received (or its own chunk at $s = 0$) to $(\text{rank} + 1) \bmod P$ and receives from $(\text{rank} - 1) \bmod P$. Total data per rank: $(P - 1) \cdot C$. Good bandwidth efficiency, high latency (linear in P).

Recursive Doubling — $\log_2 P$ rounds. At round i rank r exchanges its currently-held block of size $2^i \cdot C$ with peer $r \oplus 2^i$. Requires P to be a power of two. Data per rank: $(P - 1) \cdot C$, same as ring; however only $\log_2 P$ rounds, so latency is logarithmic.

Swing — $\log_2 P$ rounds with alternating-direction doubling peers. At round i the signed distance is $d_i = (-2)^i$ (i.e. $1, -2, 4, -8, \ldots$). Rank r sends its accumulated chunk set to $(r + d_i) \bmod P$ and receives from $(r - d_i) \bmod P$; partial-sum subsets of the distance sequence cover every residue class mod $P = 2^n$, so the algorithm finishes in exactly $\log_2 P$ rounds. The alternating sign shortens the longest logical hop versus pure recursive doubling on ring/torus fabrics while keeping the same data volume.

Note: the original Swing paper (Di Girolamo et al., 2023) uses Jacobsthal distances $[+1, -1, +3, -5, +11, \ldots]$, which are designed for AllReduce on torus topologies. Those distances do not form a spanning set for AllGather at every $P = 2^n$ (e.g. $P = 4$ yields only 3 of 4 ranks). The $d_i = (-2)^i$ variant used here preserves the swing pattern of direction alternation while guaranteeing correctness at every power-of-two P.

2.2 Broadcast

Root rank distributes one tensor to every other rank.

Binary Tree — a heap-shaped binary tree rooted at the broadcast root. Logical index $i = (\text{rank} - \text{root}) \bmod P$; parent $\lfloor (i-1)/2 \rfloor$, children $2i+1, 2i+2$. A rank forwards the data to each in-range child after receiving. Depth $= \lceil \log_2 P \rceil$, fan-out $= 2$.

Binomial Tree — $\lceil \log_2 P \rceil$ rounds. At round i every rank that already holds the data and whose logical index is $< 2^i$ sends to peer logical $+ 2^i$ if that peer exists. The tree thus doubles in size every round. Depth matches the binary tree, but the root's out-degree grows each round rather than being capped at 2.

3 Implementation notes

- `collectives.py` — algorithms themselves.
- `test_correctness.py` — spawns 2, 4, 8 ranks and checks every algorithm's output against the mathematically expected value for several roots (all tests pass).
- `run_experiments.py` — driver that launches P gloo workers via `torch.multiprocessing.spawn`, runs 1 warm-up + 3 timed iterations for every (algorithm, size, P) cell, and writes `results.json`.
- `plot_results.py` — renders the four required plots from `results.json` into `plots/`.

Timing. A barrier synchronises all ranks before the start of each iteration; each rank records `perf_counter()` deltas. The per-cell completion time is the **median** across trials of the **max across ranks** (a collective is only finished once its last participant returns).

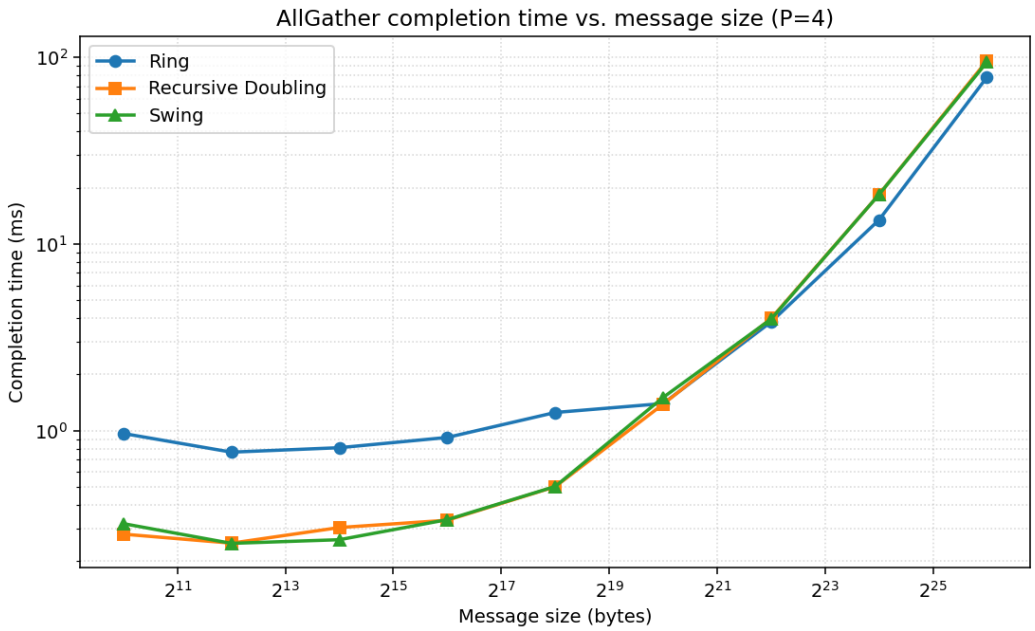
Experimental parameters.

Parameter	Value
Backend	PyTorch 2.10 · gloo, TCP over loopback (`127.0.0.1`)
Host	macOS (Darwin 25.3.0), single-host multi-process
dtype	`float32` (4 bytes/element)
Ranks tested (rank-sweep)	2, 4, 8
Message sizes tested (size-sweep)	1 KiB, 4 KiB, 16 KiB, 64 KiB, 256 KiB, 1 MiB, 4 MiB, 16 MiB, 64 MiB
Fixed P for size-sweep	4
Fixed size for rank-sweep	4 MiB
Warm-up / timed trials	1 / 3

Because all ranks live on the same machine, "message transfer" is loopback memcpy plus kernel TCP stack overhead; latency bottoms out near 150–300 μs even for trivial messages. This matters when reading the small-message end of the plots.

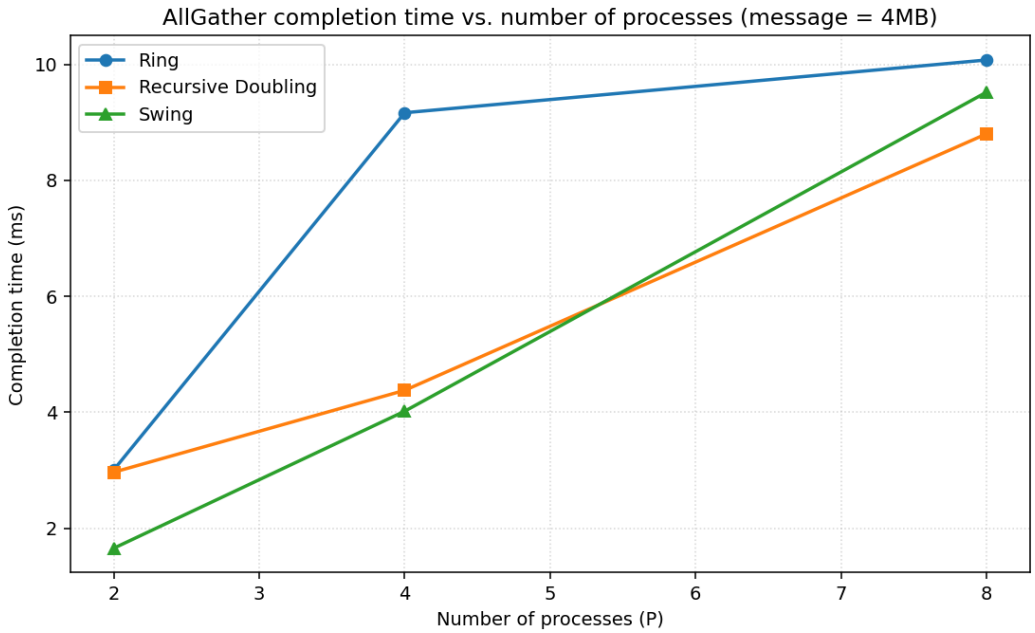
4 Results

4.1 AllGather vs. message size (P = 4)



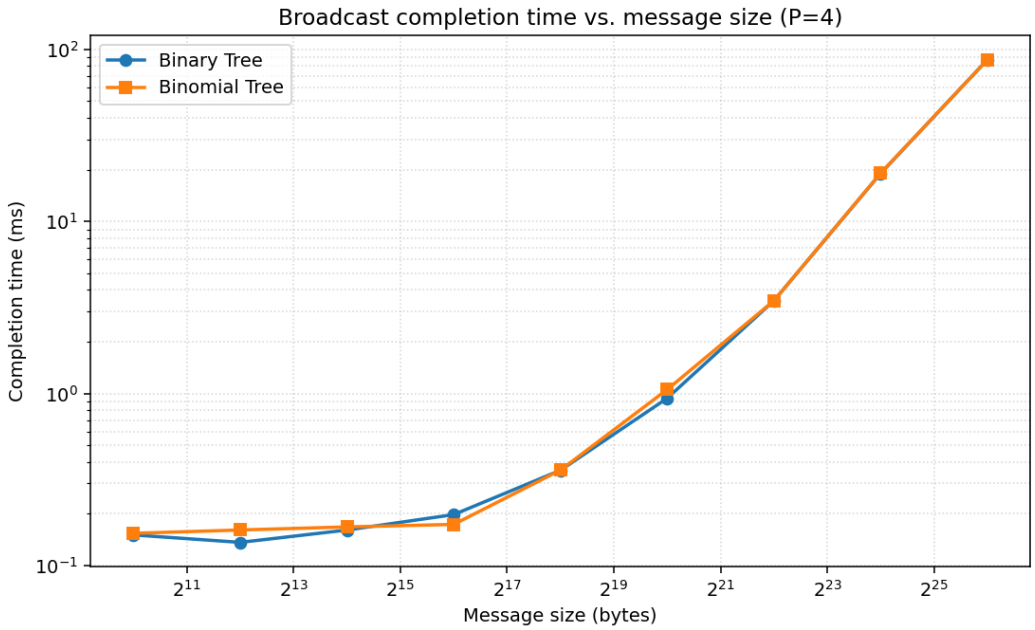
- Below ∼1 MiB the **ring** is ∼3× slower than recursive doubling or swing. With $P = 4$ the ring uses $P - 1 = 3$ sequential send/recv rounds versus $\log_2 P = 2$ for the others, and at small sizes time is dominated by per-round latency — so the ring pays the extra round.
- From ∼4 MiB up all three lines converge; every algorithm moves the same total bytes per rank, and once the per-message latency is amortised the wire time dominates.
- Above 16 MiB the **ring actually wins slightly** (∼78 ms vs. ∼94 ms at 64 MiB). Recursive doubling and swing contiguously exchange ever-larger blocks; the ring keeps each round's message at the original chunk size, which gives the NIC/TCP stack smaller, more pipelineable messages. This pipelining advantage is the classical argument for the ring at bandwidth-bound sizes.

4.2 AllGather vs. process count (message = 4 MiB)



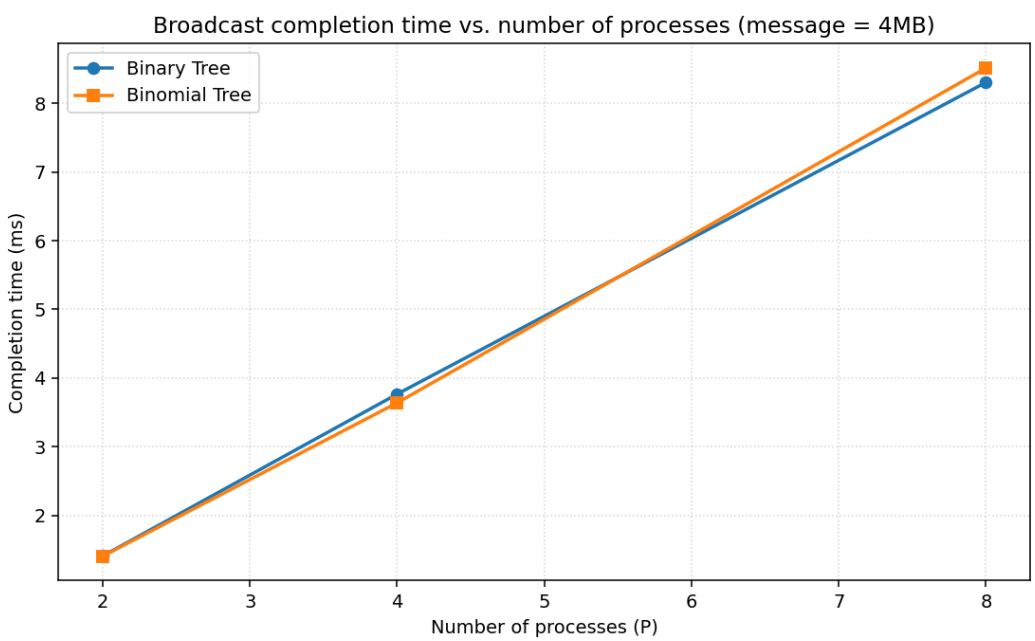
- At $P = 2$ all three algorithms do exactly one exchange; times match ($\sim$1.7–3.0 ms, variation is measurement noise).
- At $P = 4$ the ring takes 9.2 ms, $\sim$2$\times$ recursive doubling / swing, matching the $\log P$ vs. $P - 1$ round counts.
- At $P = 8$ all three converge around 9–10 ms: data per rank has grown to $(P-1) \cdot C$ regardless of algorithm and loopback bandwidth saturates.

4.3 Broadcast vs. message size (P = 4)



- Binary tree and binomial tree are within measurement noise across the whole range. At $P = 4$ both have depth 2; the binary tree's root has fan-out 2 (sends sequentially to children 1 and 2); the binomial tree's root also sends twice (at round 0 to rank 1, at round 1 to rank 2). Identical amount of serialised work at the root means identical completion time.

4.4 Broadcast vs. process count (message = 4 MiB)



- Both trees scale like $\lceil \log_2 P \rceil \cdot (\text{message}/\text{bandwidth})$ plus a constant latency per hop. $P =$

$2 \to 1$ hop, $P = 4 \to 2$ hops, $P = 8 \to 3$ hops; times grow roughly linearly with the hop count ($1.4 \to 3.7 \to 8.4$ ms).

- Again binary and binomial trees overlap — the two trees have the same depth on $P = 2^n$, and on our single-host fabric the difference in internal fan-out doesn't translate into different wall-clock because each rank's send is serialised against the others.

5 Summary

- **Small messages (latency-dominated):** log-depth algorithms (recursive doubling, swing, both trees) beat the ring on AllGather.
- **Large messages (bandwidth-dominated):** the ring catches up and can even win on AllGather thanks to smaller, pipelineable messages; tree depth stops mattering for broadcast because each hop carries the full payload.
- The two broadcast trees we compared are essentially equivalent in steady state on a flat fabric; their differences would show up on hierarchical or heterogeneous networks, which loopback is not.

6 File manifest

```
Networks Assignment 5/
|-- collectives.py         # the five algorithms
|-- test_correctness.py    # correctness tests (all pass)
|-- run_experiments.py     # experiment driver
|-- plot_results.py        # plotting
|-- results.json           # raw timing data
|-- experiment_log.txt     # stdout of run_experiments
|-- plots/
|   |-- allgather_vs_size.png
|   |-- allgather_vs_ranks.png
|   |-- broadcast_vs_size.png
|   `-- broadcast_vs_ranks.png
`-- report.md              # markdown source of this report
```

Source code and full history are available at https://github.com/arav-p/networks-assignment-5.

7 How to reproduce

```
# clone the repository
git clone https://github.com/arav-p/networks-assignment-5.git
cd networks-assignment-5

# correctness
python3 test_correctness.py

# experiments (defaults match those reported above)
python3 run_experiments.py \
    --max-bytes $((64*1024*1024)) \
    --max-ranks 8 \
    --fixed-size $((4*1024*1024)) \
```

```
    --fixed-ranks 4

# plots
python3 plot_results.py
```